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Digital Forensics

14x065

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0.1 Principles of digital forensics

1. **Acquisition:** Freeze the system (bit-wise copy, write-blocker) to avoid evidence alteration.
2. **Identification:** Extract data – recover deleted files, reveal steganography, parse metadata.
3. **Evaluation:** Reconstruct a timeline, cross-correlate events, assess reliability.
4. **Presentation:** Translate findings into layman-friendly reports or court-ready exhibits.

Typical users: criminal prosecutors, civil litigants, private corporations, insurance firms, journalists.

0.2 Digital multimedia forensics: definition, classification.

Because cheap editing tools exist, “seeing is believing” no longer holds.

We therefore need **multimedia forensics**: the science that reveals the *history* of a digital image, video or audio file by analysing its internal traces.

Media types covered: image, video, audio.

Threats: opinion manipulation, fake news, hoaxes, reputation damage.

0.2.1 Idea

Every device or processing step leaves **inherent traces**—“digital fingerprints”—that can be measured even without embedded watermarks.

Digital fingerprints

- **In-camera fingerprints:** sensor noise pattern, lens distortion, CFA interpolation, on-board compression.
 - **Out-camera fingerprints:** re-sampling, re-compression, contrast stretch, sharpening, social-platform re-encoding.
 - **Scene (geometric) fingerprints:** lighting direction, shadow consistency, chromatic aberration, specular highlights in eyes.
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0.3 Active and passive approaches for multimedia authentication.

0.3.1 Active approach

You *add* authentication data at the source.

- **Cryptographic signature**
 - Hash selected features → sign with private key → receiver verifies.
 - Can be time-stamped on a blockchain.
 - Needs PKI and trusted camera firmware.
- **Fragile / semi-fragile watermarking**
 - Embed an imperceptible mark that breaks (fragile) or survives benign change (semi-fragile).
 - Requires embedding hardware/software; marks may be removed by re-encoding.

Disadvantages

- Crypto alone does *not* say *what* was changed, only *that* something changed.
 - Bit-level integrity is too strict for lossy compression.
 - Protection disappears after decryption.
 - Watermarks degrade quality and are vulnerable to collusion.
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0.3.2 Passive and blind approach: multimedia forensics

No extra data added.

From the *received* file only, we answer:

- **Source identification:** which brand / model / individual device?
- **Integrity verification:** was the file recompressed, cropped, spliced?

Advantages

- Works on legacy files.
- No standards to negotiate, no embedding hardware, no key management.
- Compatible with social-media re-distribution chains.

0.4 Image generation pipeline

1. **Light projection** – photons hit the sensor mosaic.
2. **Charge $\rightarrow$ voltage $\rightarrow$ digital value** – amplifier + ADC.
3. **White-balance gain** – scale R, G, B channels.
4. **Color Filter Array (CFA) demosaicking** – interpolate missing colours.
5. **Color & gamma correction** – map to sRGB.
6. **Sharpening / denoising filters.**
7. **JPEG compression & storage.**

Each step injects device-specific or software-specific traces.

0.5 Camera identification based on photo response non-uniformity (PRNU)

Sensor pixels respond slightly differently to the same light.

The dominant component of this **pixel non-uniformity** is **PRNU** – a multiplicative noise pattern unique to each sensor.

Model:

$$I(x, y) = I_0(x, y) + I_0(x, y) K(x, y) + N(x, y)$$

where

- I_0 – ideal noise-free image,
- K – PRNU factor ($\approx 0.5\%$ – 2% amplitude),
- N – additive noise.

Estimation: take 30–50 flat-field or natural images from the camera, compute noise residuals, average them to obtain the **reference PRNU pattern** $\hat{K}$.

Verification: extract noise from a suspect image, correlate with $\hat{K}$; peak $>$ threshold same camera.

0.6 Media forensics based on in-camera fingerprints.

0.6.1 Source identification

- **Device class:** DSLR vs phone vs scanner.
- **Brand / model:** CFA layout, JPEG quantization tables, internal sharpening kernel.
- **Individual unit:** PRNU, defective-pixel cluster.

0.6.2 Usage of digital fingerprints

- **Source ID:** compare extracted fingerprint against database of reference patterns.
- **Tamper detection:** local absence or mismatch of fingerprint splice; presence of *external* fingerprint insertion from another device.

Examples

- Bayer CFA interpolation introduces periodic correlations → detectable.
- Different vendors use different JPEG Q-matrices → quick model lookup.
- Scanner noise vs camera PRNU → separate CG vs real.

0.6.3 Tampering detection

A pasted region often comes from a different camera or undergoes re-compression → CFA or PRNU inconsistency.

0.7 Application of geometrical fingerprints in digital forensics.

0.7.1 Copy-move

Part of the image is cloned to hide something.

Detection: divide image into overlapping blocks, extract features (DCT, Zernike, BRISK), look for unusually similar blocks in the *same* image.

Limitation: cannot tell which copy came first.

0.7.2 Highlights

Eye corneas act like little mirrors.

Fit a sphere to specular highlights → estimate light direction.

Multiple inconsistent highlight directions → composite.

0.7.3 Shadows

Under Lambertian+single-point-light assumptions, shadow boundaries constrain the 3-D light position.

Intersect constraints from several objects; empty intersection → fake.

0.7.4 Chromatic aberration

Lenses bend wavelengths differently → coloured fringes.

Cut-and-paste often breaks fringe continuity → detect splice.

0.8 Out-camera fingerprints in digital forensics.

Post-production steps leave their own marks:

- **Re-compression** – double-quantization periodic histograms.
 - **Re-sampling** – periodic linear dependencies detectable by EM or CNN.
 - **Contrast stretch** – histogram gaps/peaks.
 - **Median filtering** – destroys Gaussian noise tail, introduces edge “cross-shaped” artifacts.
 - **Sharpening** – overshoot along edges (USM kernel).
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0.9 Principles of image JPEG compression.

JPEG = lossy codec tuned for natural photos.

Exploits **two redundancies**:

1. **Statistical redundancy** – spatial correlation (DCT).
2. **Psychovisual redundancy** – humans ignore high frequencies (quantization).

Extra leverage: eye sees luminance (Y) better than chrominance (C_b, C_r) → down-sample colour 2×.

0.10 JPEG compression: general compression scheme

Encoder pipeline:

T (transform, lossless) $\rightarrow Q$ (quantizer, lossy) $\rightarrow C$ (entropy coder, lossless).

Removing Q gives lossless mode, but compression ratio $\approx 2:1$ only.

0.10.1 Details

1. Split Y, C_b, C_r into 8×8 blocks.

2. **DCT** each block:

$$F(u, v) = \text{DCT}(f(x, y)).$$

3. **Quantize**:

$$F_q(u, v) = \text{round}(F(u, v)/Q(u, v)).$$

Larger $Q(u, v)$ at high frequencies $\rightarrow$ aggressive compression.

Q matrices scaled by **Quality Factor** $QF \in [1, 100]$.

4. **Zig-zag scan** $\rightarrow$ 1-D sequence.

5. **DPCM** on DC coefficient, **RLE** on AC run-lengths, **Huffman** / **arithmetic coding**.

0.11 JPEG compression: sources of redundancies

- **Spatial** – neighbouring pixels similar.
- **Spectral** – R, G, B highly correlated (why we convert to YC_bC_r).
- **Temporal** – successive frames alike (used in Motion-JPEG).

Irrelevance counterparts:

eye tolerates blur, colour error, and temporal flicker – all quantized more coarsely.

0.12 JPEG compression: particularities of luminance and chrominance channels processing

RGB $\rightarrow$ YC_bC_r (linear):

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.5 \\ 0.5 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Chroma subsampling 4:2:0 $\rightarrow$ keep Y at full resolution, C_b, C_r at half.
Data volume halved with almost no visual loss.

0.13 JPEG compression artefacts

Low QF introduces:

- **Blocking** – 8»8 grid discontinuities.
- **Ringing** – Gibbs phenomenon near edges.
- **Blurring** – high frequencies zeroed.
- **Graininess** – coarse quantisation noise.

All leave **detectable statistical footprints**.

0.14 Detection of double JPEG compression

An image re-saved with **different JPEG settings** acquires **double-quantisation (DQ) footprints** in DCT histograms.

Why care? DQ file was modified after original export possible forgery.

0.14.1 Single quantisation (SQ)

$$y = \text{round}(x/a)$$

x recovered by $x' = a y$ (lossy).

0.14.2 Double quantisation (DQ)

$$z = \text{round}(\text{round}(x/a) \cdot a/b)$$

- If $a < b \rightarrow$ histogram shows **empty bins** (holes).
- If $a > b \rightarrow$ **periodic peaks**.

Detection: compute Fourier transform of DCT-coeff histogram; strong peaks at $ka/\text{gcd}(a, b)$ DQ.

Reliable when $QF_1 < QF_2$.

Benford / First-Significant-Digit (FSD) test: single JPEG obeys Benford's law; double compression violates it another detector.

0.15 Image editing forensics: inconsistency in light

Copy-paste usually merges objects lit by **different sources**.

Under simple assumptions (Lambertian surface, distant light, uniform albedo) the **light direction** can be estimated from the **intensity gradient** on an occluding contour.

Different directions splice.

0.16 Image editing forensics: local filtering traces

Median filtering erases noise spikes and JPEG blockiness; output histogram becomes **piece-wise constant** – unnatural for natural images.

Detect by checking **histogram bin clusters** or by CNN.

Unsharp masking subtracts blurred version:

$$g = f + \lambda(f - f_{\text{blur}})$$

creates **overshoot artefacts** (dark/light parallel stripes) along edges $\rightarrow$ kernel fingerprints detectable.

0.17 Photo manipulation with inconsistent shadows: basic assumptions

- Only **one dominant point light** (sun, flash).

- Two shadow classes:
 - **Cast shadow** – projected on ground.
 - **Attached shadow** – on object itself (self-shadow).
- No geometry model needed; only **2-D projection constraints**.

For n objects draw n half-planes / wedges; intersection must be **non-empty**.
 Empty set inconsistent tampering.

0.18 Photo manipulation with inconsistent shadows: cast shadow

0.19 Photo manipulation with inconsistent shadows: attached shadow

Attached shadow edge gives **gradient direction** half-plane constraint.
 User can adjust auto-detected normal n .

0.20 Photo manipulation with inconsistent shadows: mathematical model

A 2-D line through point $p = (p_x, p_y)$ with normal $n = (n_x, n_y)$:

$$n_x x + n_y y - (n_x p_x + n_y p_y) = 0$$

Half-plane (attached shadow):

$$n_x x + n_y y \geq n_x p_x + n_y p_y$$

Cast shadow uses **two** such lines (umbra & penumbra).
 Stack all inequalities: $Nx \geq P$.

If system **infeasible**, add **slack variables** $s \geq 0$:

$$Nx - P + s \geq 0$$

Solve **linear program**; minimal sum of slacks > 0 inconsistent shadows forgery.
Sign ambiguity: light could be behind camera solve **both**
 $Nx - P \geq -s$ and $-Nx - P \geq -s$; keep the feasible one.

0.21 Traces of image editing: resampling detection

Splicing often needs **scale / rotate** → **re-sampling interpolation** introduces **periodic linear dependencies** among pixels.

Model: $y = Hx$ with H the interpolation matrix.

For large images H becomes rank-deficient **periodic predictability**.

Compute **probability map** of pixel being linearly predicted; Fourier transform shows **harmonic peaks** alerts to geometric tampering.

0.22 Traces of image editing: detection of global contrast enhancement

Any global tone curve **expands dynamic range** → **energy grows**, especially in **high DFT frequencies**.

Natural images have **low-pass DFT**; isolated **HF peaks** contrast mapping applied.

Histogram may also show **isolated spikes / gaps**.

0.23 Traces of image editing: detection of local contrast enhancement

Local enhancement (e.g. brushed in Photoshop) only inside spliced region.

Sliding window: divide image into blocks ($200 \times 200 \dots 20 \times 20$), apply global detector to each.

Map of positives pinpoints tampered area at block resolution.

Run on **R, G, B** separately for better localisation.